UMA MALANI

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EDUCATION

The University of Chicago Chicago, IL

Bachelor of Science in Computer Science and Statistics, expected June 2029 | GPA: 3.94/4.0

Relevant Coursework: Calculus III, Pathways in Data Science (machine learning), Computer Systems Programming; *In progress (Fall 2026):* Linear Algebra, Discrete Mathematics, Computer Systems Programming II.

Honors: Dean's List (2025-2026).

The University of Chicago Laboratory High School Chicago, IL

Diploma, May 2025 | Honors: AP Scholar, National Merit Commended Scholar, Illinois State Scholar, AP Statistics (5).

TECHNICAL & QUANTITATIVE EXPERIENCE

Jump Trading *Quantitative Research Intern* Jun 2024 – Aug 2024

- Fine-tuned GPT-3 on financial news headlines to classify sentiment and forecast asset price direction.
- Applied classification and regression models to real-world market data to predict asset price movements.
- Unified market data with model predictions into a single dataset, achieving ~80% accuracy on market-direction forecasting.

University of Chicago Human-Robot Interaction Lab *Research Assistant* Jun 2025 – Jun 2026

- Designed a theoretical framework for robot dialogue incorporating self-disclosure and empathetic listening that guides ongoing interaction design.
- Implemented the dialogue system for the Temi robot in Kotlin, translating the framework into deployable interaction scripts.

Actelligent *Data Science Intern* Jun 2026 – Aug 2026

- Built Python NLP pipelines that extract themes (BERTopic) and sentiment (VADER/FinBERT) from earnings-call transcripts across ~9,000 global companies on AWS, converting the output into sector-neutral long/short equity portfolios.
- Backtested and validated the resulting signals against company fundamentals using out-of-sample, random-control, and transaction-cost-adjusted methods.

ADDITIONAL EXPERIENCE

Widget Factory (YC-backed startup) *Quality Assurance Engineer (Part-Time)* Mar 2026 – Present

- Develop and execute test plans for product features and write detailed, reproducible bug reports for the engineering team.

Computer Science Instructional Laboratory (CSIL) *Staff Member* Spring 2026 – Present

- Build and maintain internal web tools for CSIL staff, including a checkout system, shift scheduler, and alumni database.
- Image and configure lab computers for specialized use cases, including sandboxed exam environments for professors.

LEADERSHIP & ACTIVITIES

Paragon Global Investments *Quantitative Analyst* Feb 2026 – Present

- Researched, built, and pitched "Residual Industry Momentum," a factor-residualized long-short strategy over 49 industry portfolios (1967-2026); isolated a statistically significant 4.4% annualized alpha ($t = 2.15$) by stripping Fama-French 3-factor exposures, halving momentum-factor loading and raising the Sharpe ratio versus the baseline.
- Backtested 21 strategy variants in Python across momentum lookbacks, residualization windows, and factor models, evaluating alpha via Newey-West (HAC) time-series regressions.

Data Science Society *Presenter & Analyst* Fall 2025 – Present

- Analyzed 200+ podcasts, articles, and missing-persons reports to test the "missing white woman" hypothesis, surfacing coverage disparities by race and gender; deliver weekly educational talks.

Girls Who Code *Co-founder & President* May 2022 – Apr 2025

- Founded the chapter, designed a weekly curriculum, and led teams in the ACSL Competition and Technovation Challenge.

Science Olympiad *Varsity Member* Nov 2022 – Apr 2025

- State-level competitor: 4th overall (2023), 3rd overall (2024–2025).

SKILLS

Programming: Python, R, SQL, Kotlin, JavaScript, C, HTML/CSS.

Libraries & Tools: pandas, NumPy, scikit-learn, statsmodels, PyTorch, BERTopic, FinBERT, AWS, React, Jupyter, Git.

Languages: Conversational Mandarin.